

Education

Indian Institute of Technology Bombay, India

Master of Science in Applied Statistics and Informatics

July 2024 - Present

Birla Institute of Technology Mesra Ranchi, India

Bachelor of Science (Exit From IMSc)

Dec 2021 -May 2024

Summary

AI/ML Engineer with a strong foundation in Applied Statistics from IIT Bombay, specializing in machine learning, NLP, and Generative AI systems. Experienced in building end-to-end pipelines, including LLM-based and data-driven solutions, to solve complex real-world problems at scale.

Professional Experience

Data Engineering Intern (Financial Data Processing and Analytics)

Goa Institute of Management, India | Supervisor: Prof. Abhinav Sharma

May 2025 – June 2025

Goa, India

- **Objective:** Build a unified dataset to analyze relationships between social sentiment and firm breach events
- **Methodology:** Engineered data pipeline using web scraping + APIs, NER, fuzzy matching, & data normalization for cross-source entity resolution
- **Outcome:** Enabled high-quality dataset integration and improved linkage accuracy for large-scale sentiment and risk analysis

Publication: Paper Under Review in Egyptian Informatics Journal

LLM-based Emotion Mining on Students' Qualitative Feedback Data Enriching Institutional Policy Building

Dr. Pawan Kumar Tiwari (BIT Mesra) | Dr. Suman Pandey (GIST, South Korea) | Aug 2024 - July 2025

- Converted unstructured qualitative feedback from PDF documents into clean CSV files using automated extraction with PyMuPDF, enabling reliable analysis, and reducing manual data handling across large collections of feedback files.
- Represented feedback text as sentence embeddings, applied UMAP to reduce dimensions to 100, and used K-Means clustering to group similar responses, helping to identify common patterns and recurring topics in user feedback.
- Tuned clustering parameters and evaluated results using the Silhouette score (0.95) and the Davies–Bouldin index (0.15) to confirm strong separation between groups and a consistent structure within clusters.
- Analyzed the sentiment and emotional tone of feedback using transformer-based language models, allowing for the comparison of positive and negative responses, and supporting a data-driven assessment of faculty performance trends.
- Assigned faculty members to appropriate roles by comparing the semantic similarity of feedback embeddings and applying the Hungarian algorithm to find optimal matches, improving alignment between roles and feedback-derived strengths.

Projects

Modeling and Forecasting Green Taxi Demand | Prof. Monika Bhattacharjee, IIT Bombay | [GitHub](#)

- **Objective:** Develop a high-accuracy predictive model for taxi demand to optimize urban mobility and resource allocation
- **Methodology:** Processed 1.2M+ trip records, integrated external weather features, and implemented XGBoost and Linear Regression with advanced feature engineering and data preprocessing
- **Outcome:** Achieved $R^2 = 0.94$, significantly improving prediction performance and model reliability
- **Future Improvement:** Extend with deep learning models (LSTM/Transformer) and real-time forecasting pipelines

Product Review Rating Prediction using NLP | [GitHub](#)

- **Objective:** Improve reliability of product ratings using NLP-based sentiment analysis
- **Methodology:** Processed 28K reviews, used MiniLM embeddings + UMAP + K-Means + RoBERTa for sentiment scoring
- **Outcome:** Generated weighted sentiment scores outperforming traditional star ratings
- **Future Improvement:** Deploy as an API for real-time review scoring in e-commerce platforms

RAG-Powered DSA Assistant | [GitHub](#)

- **Objective:** Build an intelligent assistant for accurate DSA concept retrieval and explanation
- **Methodology:** Developed RAG pipeline using embeddings + vector DB + LangChain, optimized chunking and retrieval
- **Outcome:** Reduced hallucination and improved answer accuracy for interview-level queries
- **Future Improvement:** Add multi-agent reasoning and adaptive learning from user queries

Multi-Agent Resume Evaluation System | [GitHub](#)

- **Objective:** Build a scalable AI system for semantic resume–job alignment and automated candidate evaluation
- **Methodology:** Architected an end-to-end pipeline with document parsing, structured information extraction, embedding-based similarity scoring, and LLM orchestration (prompt engineering + controlled generation)
- **Outcome:** Delivered explainable role-fit scoring with section-level insights and automated feedback generation, improving decision efficiency
- **Future Improvement:** Extend with retrieval-augmented generation (RAG), and fine-tuned LLMs for domain-specific optimization

Technical Skills

Programming & Libraries: Python, SQL, Java, R, Pandas, NumPy, Scikit-learn, Matplotlib

AI / Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Cross Validation

Generative AI & NLP: Transformers, BERT, Text Embeddings, spaCy, NLTK, TensorFlow, Prompt Engineering, Fine-Tuning, RAG

Agentic AI & Systems: LangChain, LangGraph, AutoGen, Multi-Agent Systems, Tool Calling, Orchestration, Vector Databases

Data Science & Analytics: EDA, Statistical Modeling, Hypothesis Testing, Time Series Analysis

Relevant Coursework and Certifications

AI & Data Science | Algorithms | Computational Statistics | R Programming | Time Series Analysis | Regression Analysis | Design of Experiments | Deep Learning (Scaler) | SQL (HackerRank) | Java Programming (Udemy) | 160 Days of Problem Solving – GfG | Bloomberg Finance Fundamentals

Hobbies

Playing Cricket | Gym & Fitness | Reading Research Papers & Tech Blogs | Building AI/ML Projects | Problem Solving